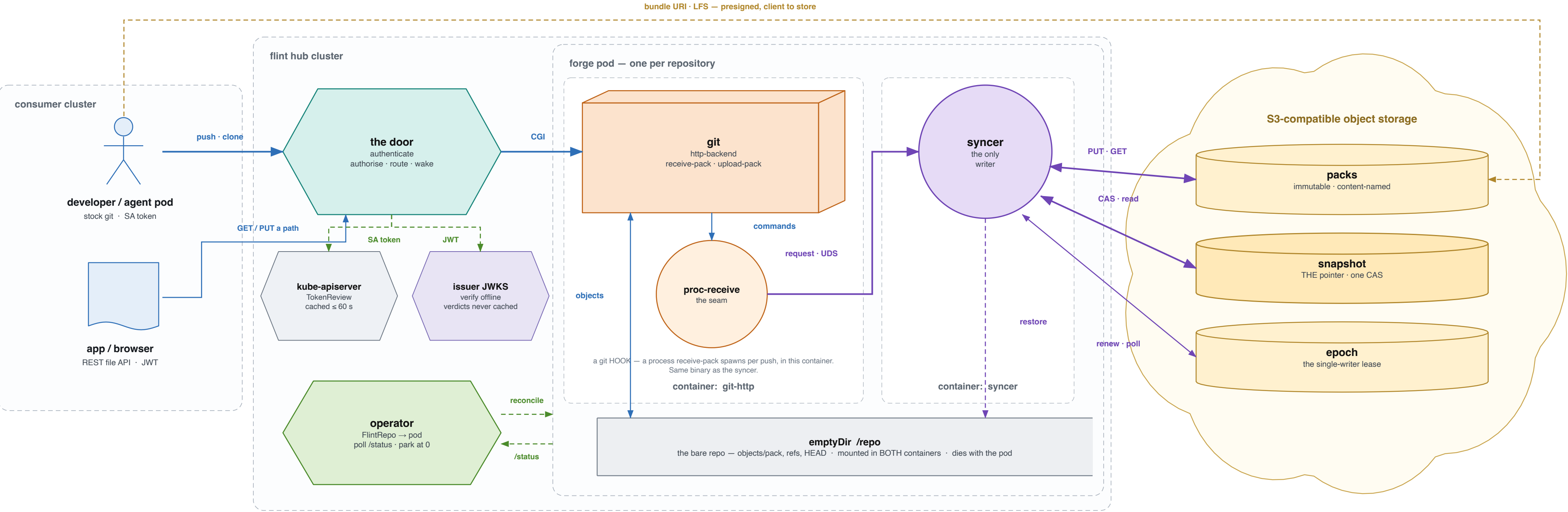


# Data flow

Stock git serves every byte. One syncer beside it is the only process that can write S3.



TWO CREDENTIAL KINDS, SERVED AT ONCE. The door routes on the token's iss: a pod's ServiceAccount token goes to TokenReview at the apiserver, an issuer-minted JWT to offline signature verification against the cached JWKS. A JWT's sub — a person — becomes the principal for spec.consumers, for the branch policy, and as the commit author.

AND THE DOOR REFUSES TO START if the cluster's ServiceAccount issuer and the configured JWT issuer are the same string: every pod token would then route to the offline verifier, which cannot know a pod was deleted.

ONE CAS ON THE SNAPSHOT is the transaction boundary. Told “ok” means that object moved; a 412 on it is a fence, not a retry.

→ data plane — every byte of a clone or push      → durable path — the only writer of the bucket      → control plane — never reads an object      → bypass — the pod sees a few hundred bytes

## What the abbreviations mean

|                                                                                                         |                                                                                                        |                                                                                                            |                                                                                                        |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>CAS</b><br>compare-and-swap — a write that lands only if the object is still the version you read    | <b>ETag / If-Match</b><br>S3's precondition: the ETag you read, offered back as the terms of the write | <b>fence</b><br>a lost CAS. Another server holds this repository, so this one stops READING too, and exits | <b>CGI</b><br>Common Gateway Interface — how the runner invokes git http-backend, once per request     |
| <b>UDS</b><br>Unix domain socket — the hook-to-syncer channel, inside the pod, never over a network     | <b>proc-receive</b><br>the git hook that lets the server, not git, decide each ref update              | <b>pre-receive</b><br>the git hook that applies the branch policy, and can refuse the whole push           | <b>emptyDir</b><br>a Kubernetes volume that lives and dies with the POD — not with a container         |
| <b>SA token</b><br>a Kubernetes ServiceAccount token, projected into the pod and rotated by the kubelet | <b>TokenReview</b><br>the Kubernetes API that says whether a ServiceAccount token is still valid       | <b>JWT</b><br>JSON Web Token — a signed bearer token, verified at the door with no round trip              | <b>JWKS</b><br>JSON Web Key Set — the issuer's signing keys. The KEYS are cached, the verdicts are not |
| <b>iss / sub / exp</b><br>a JWT's issuer, its subject — the principal — and its expiry                  | <b>LFS</b><br>Git Large File Storage — big blobs kept beside the repository, moved by presigned URL    | <b>bundle URI</b><br>git's own opt-in: the server names a pack the client fetches straight from S3         | <b>FlintRepo</b><br>the custom resource that one forge repository is declared as                       |